



STEM SMART Single Maths 27 - Binomial Expansion - Natural n >

Binomial Expansion - Natural n

A-level Maths Topic Summaries - Series

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P1

Fill in the blanks below to complete the notes on binomial expansion for integer powers of n .

The binomial theorem is used to expand brackets of the form $(a + b)^n$. For integer powers of n ,

$$(a + b)^n = \binom{n}{0} \square + \binom{n}{1} \square b + \binom{n}{2} a^{n-2} \square + \dots + \binom{n}{n-1} a \square + \binom{n}{n} b^n$$

The $\binom{n}{r}$ brackets are the numbers in triangle. It is useful to know that

$$\binom{n}{0} = \binom{n}{n} = \square$$

$$\binom{n}{1} = \binom{n}{n-1} = \square$$

Items:

1
a
 a^{n-1}
 a^n
 b^2
 b^{n-1}
n
Pascal's

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Expand and Simplify Binomials

Pre-Uni Maths for Sciences C1.1

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P1

Part A

$$(x + 1)^4$$

Expand and simplify $(x + 1)^4$.

The following symbols may be useful: x

Part B

$$(z + 2a)^3$$

Expand and simplify $(z + 2a)^3$.

The following symbols may be useful: a, z

Part C

$$(a - b)^5$$

Expand and simplify $(a - b)^5$.

The following symbols may be useful: a, b

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Binomial Expansion 3

Pre-Uni Maths for Sciences C1.7

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level C1

Expand $(3 - a)^4$ in ascending powers of a up to and including the term in a^3 . Hence, without using a calculator, evaluate $(2.9)^4$ correct to 2 decimal places

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Binomial Expansion 7

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P2

Use binomial expansion to expand and simplify the expression $\left(ax^2 + \frac{2}{x}\right)^5$.

The following symbols may be useful: x

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Find Coefficients 1

Pre-Uni Maths for Sciences C1.2

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P1

Find the coefficient of x^3 in the expansion of:

Part A

$$(x - 10)^5$$

$$(x - 10)^5$$

Part B

$$(2x - \frac{1}{2})^6$$

$$\left(2x - \frac{1}{2}\right)^6$$

Part C

$$(x - y)^{10}$$

$$(x - y)^{10}$$

The following symbols may be useful: x, y

Part D

$$\left(x - \frac{1}{x}\right)^7$$

$$\left(x - \frac{1}{x}\right)^7$$

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Find Coefficients 2

Pre-Uni Maths for Sciences C1.3

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P1

Without expanding the binomials, find:

Part A

Coefficient of x^4y^6

The coefficient of x^4y^6 in the expansion of $(x^2 + 3y^2)^5$.

Part B

Coefficient of x^{20}

The coefficient of x^{20} in the expansion of $(x^2 + 3x)^{12}$.

Part C

The coefficient of ab^7

The coefficient of ab^7 in the expansion of $(a + \frac{1}{4}b)^8$.

Part D

Constant term

The constant term in the expansion of $\left(\frac{x^2}{2} - \frac{8}{x}\right)^9$.

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Group and Expand

Pre-Uni Maths for Sciences C1.5

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P1

Expand $(1 - 2x + 3x^2)^7$ in ascending powers of x as far as x^3 .

The following symbols may be useful: x

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Binomial Theorem Applied to Variable Acceleration

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P2

The force $F(t)$ N on a particle of mass 4 kg at time t s is given by

$$F(t) = (\sqrt{t} + 2)^5 - (\sqrt{t} - 2)^5$$

for $t \geq 0$.

At $t = 0$ the particle is at rest at the origin. Find an expression for its displacement from the origin, x m, in terms of t .

The following symbols may be useful: t , x

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